

Diego Núñez Simms

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ENVIRONMENTAL DATA SCIENTIST & AGRONOMIST

Data Science — Bioclimatic Modeling — GIS & Remote Sensing — Agrometeorology

PROFESSIONAL SUMMARY

Agronomist and environmental data scientist with 2+ years of applied experience integrating machine learning, bioclimatic modeling, and geoprocessing with physical fieldwork. Proven track record designing data pipelines (QA/QC, gap-filling) for regional weather stations and global climate models (ERA5-Land), managing ecological datasets (carbon fluxes, soil carbon, water balance), and deploying agrometeorological sensor networks. Skilled at translating complex agricultural and climatological data into actionable field insights and robust software solutions.

CORE EXPERTISE

- Environmental Data Science & Machine Learning (Python, R)
- Bioclimatic & Phenological Modeling (GDD, Chill Portions, Phenology)
- Remote Sensing & GIS (Google Earth Engine, Sentinel, Landsat, QGIS)
- Climate Reanalysis & QA/QC Pipelines (ERA5-Land)
- Ecohydrology & Carbon Flux Monitoring (Eddy Covariance, NEE, Soil Carbon)
- Agronomy & Field Operations (Calicatas, Soil Surveying, Sensor Networks)
- Software Engineering for Science (Pytest, YAML, Modular Design)

PROFESSIONAL EXPERIENCE

Research Assistant – Data Science & Agronomy

Viña Concha y Toro

2026 – Present

- Developed bioclimatic modeling pipelines (Vendimia 5.0) predicting phenological stages, cold portions, and berry maturity.
- Optimized automated QA/QC check pipelines to validate reanalysis data against physical weather station networks.
- Built Gradio and static HTML dashboards for real-time visualization of climate metrics.

MSc Researcher – Ecohydrology & Carbon Dynamics

IEB Chile & University of Chile

2024 – 2026

- Modeled Net Ecosystem Exchange (NEE) carbon fluxes using SVR and Random Forest to assess carbon sink capacity.
- Managed meteorological data workflows (quality control, gap-filling, flux partitioning) for Eddy Covariance towers.

- Authored peer-reviewed manuscript on arid shrublands functioning as a net carbon sink.
- Title of thesis: "Water and carbon dynamics in a semi-arid shrubland ecosystem."

Environmental & Spatial Data Consultant

GeoTerra.cl

2023 – 2025

- Built spatial data models and scripts (Python, R, GIS) for territorial planning and agricultural site assessment.
- Integrated multi-spectral satellite imagery (NDVI, EVI, NDWI) to track vegetation health and drought indices.

Research Intern – Agrometeorology & Soil Science

University of Chile

2024

- Monitored soil organic carbon, physical structural indices (calicatas), and GHG emissions (CO₂, N₂O) in field crops.
- Installed, calibrated, and maintained meteorological stations, soil moisture sensors, and phenology cameras.

HONORS & RECONCILIATIONS

- 1st Place in the National Soil Classification and Description Contest (Individual category).
- Graduated with High Distinction (B.Sc. and M.Sc.), University of Chile.

EDUCATION

M.Sc. in Territorial Management of Natural Resources

University of Chile

2024 – 2026

B.Sc. in Agricultural Sciences (Ingeniería Agronómica)

University of Chile

2020 – 2024

TECHNICAL SKILLS

- Programming: Python (Pandas, NumPy, Scikit-learn, PySR, joblib), R
- GIS & Remote Sensing: QGIS, ArcGIS, Google Earth Engine, Sentinel, MODIS, Landsat
- Climate Data: ERA5, reanalysis datasets, NetCDF, Grib
- Lab & Field: Eddy Covariance, GHG chamber sampling, soil carbon analysis, sensor calibration

LANGUAGES

Spanish (Native) — English (Professional Working Proficiency)